R13

Code No: 111AH

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech I Year Examinations, December - 2017
ENGINEERING DRAWING
(Common to CSE, BME, MIE, PTM)

Time: 3 hours **Max Marks: 75**

Answer all five questions
All questions carry equal marks
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1.a) The distance between Parry's corner and Egmore is 2.5 km. On inspection of the road map, its equivalent distance measures 5 cm. Draw a diagonal scale to read 50m minimum. Show on it a distance of 6350 m.

b) Construct a hyperbola, when the distance of the focus from the directrix is 65mm, an eccentricity is 3/2. [8+7]

OR

2. A fixed point F is 7.5cm from a fixed straight line. Draw the locus of a point P moving in such a way that its distance from the fixed straight line is 2/3 times its distance from F. Plot at least 9 points. Name the curve. Also draw a normal and a tangent to the curve at a point on it 6cm from F. [15]

3.a) A line PQ 50mm long is perpendicular to HP and 15 mm in front of VP. The end P nearer to HP is 20 mm above it. Draw the projections of the line.

b) A straight line PQ 10cm makes 45^0 to HP and 30^0 to VP. The end P is 3.6 cm above HP and 4 cm in front of VP. Draw the top and front views of the line. Measure the distance between the projectors of the ends of the line. [7+8]

OR

4. A semi circular plate of 80 mm diameter has its straight edge in the VP and inclined at 60^0 to the HP, the surface of the plate makes an angle of 30^0 with the VP. Draw its projections. [15]

5. A pentagonal prism is resting on one of the corners of its base on the H.P. The longer edge containing that corner is inclined at 45^0 to the H.P. The axis of the prism makes an angle of 30^0 to the V.P. Draw the projections of the solid. [15]

OR

6. A cone, base 75 mm diameter and axis 80 mm long is resting on its base on the HP. It is cut by a section plane perpendicular to the VP, inclined at 45^0 to the HP and cutting the axis at a point 35 mm from the apex. Draw its front view, sectional top view and true shape of the section. [15]

7. A right circular cone of base 60 mm diameter and 60 mm height stands vertically with its base on HP. A semicircular hole of 36 mm diameter is cut through the cone such that the axis of the hole is parallel to HP, perpendicular to VP and intersecting the axis of the cone 20 mm above the base. The flat surface of the hole is parallel to HP and perpendicular to VP. Draw the development of the lateral surface of the cone with the hole. [15]

OR

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8. A square prism of base side 60 mm rests on one of its ends on the HP with the base sides equally inclined to the VP. It is penetrated fully by another square prism of base side 45 mm with the base side equally inclined to the HP. The axes intersect at right angles. The axis of the penetrating prism is parallel to both the HP and the VP. Draw the projections of the prisms and show the lines of intersection. [15]

9. Draw the isometric view of the object whose orthographic projections are given in figure 1. All dimensions are in mm. [15]

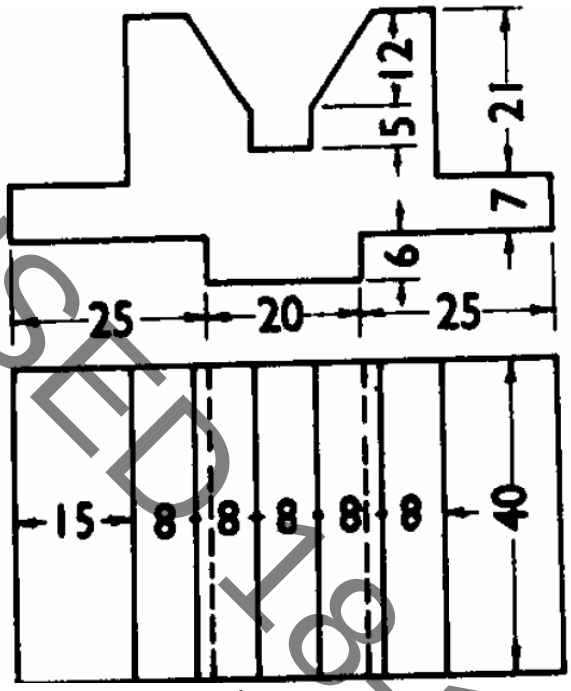


Figure: 1

OR

10.a) Draw the following views of the angular block given in figure 2.
i) Front view ii) Top view and iii) Side view. All dimensions are in mm.

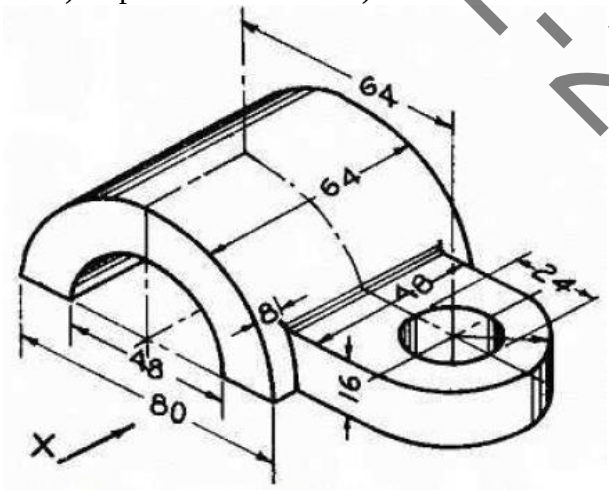


Figure: 2

b) A square lamina of 40mm side lies on the ground plane. One of its corners is touching the PP and an edge inclined at 60^0 to the PP. The station point is 40 mm in front of PP, 60 mm above the ground plane and lies in a central plane which is at a distance of 35 mm to the right of the corner touching the PP. Draw perspective view of the lamina. [8+7]

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